

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Summer Examination – 2025

Course: B. Tech

Branch: Electrical /Electrical Engineering(Electrical and Power)

Subject Code & Name: BTEEC601_Y19 Control System Engineering

Semester : VI

Max Marks: 60

Date:30/7/2025

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level /CO)	Mark
Q. 1	Objective type questions. (Compulsory Question)		12
1	The use of signal flow graph is to find which of the following	CO1	1
	a. Transfer function of control system b. Poles of control system c. Controllability of control system d. Stability of control system		
2	Identify the correct for the transfer function of positive feedback	CO1	1
	a. $\frac{G(s)}{1-G(s)H(s)}$ b. $\frac{G(s)}{1+G(s)H(s)}$ c. $\frac{G(s)}{(s)H(s)}$ d. $\frac{G(s)}{G(s)H(s)-1}$		
3	The steady state error is determined as the difference between the reference input and system output at	CO3	1
	a. $t=t_p$ b. $t=\infty$ c. $t=0$ d. $t=$ time constant		
4	The steady state error is specified in terms of	CO3	1
	a. Damping factor b. Bandwidth c. Error constants d. Speed of response		
5	For the system $\frac{C(s)}{R(s)} = \frac{16}{s^2+8s+16}$ The nature of the time response will be	CO2	1
	a. Overdamped b. underdamped c. critically damped d. None of these		
6	The roots of the characteristic equation are same as	CO2	1
	a. closed-loop zeros b. closed-loop poles c. open-loop zeros d. open-loop poles		
7	The Bode plot is obtained using	CO4	1

	a. characteristics equation	b. open loop transfer function	c. closed loop transfer function	d. overall transfer function		
8	A second order control system with $\zeta = 0$ is always				CO4	1
	a. marginally stable	b. stable	c. unstable	d. inherently stable		
9	The Routh-Hurwitz criterion gives				CO3	1
	a. relative stability	b. absolute stability	c. gain margin	d. phase margin		
10	The analysis of multiple-input multiple-output system is conveniently studied by				C06	1
	a. state space approach	b. root locus approach	c. characteristic equation approach	d. Nichols chart		
11	The state transition matrix $\phi(t)$ is given by				CO6	1
	a. $[sI] - [A]$	b. $\{[sI]-[A]\}^{-1}$	c. $L^{-1}\{[sI]-[A]\}^{-1}$	d. $L^{-1}\{[sI]-[A]\}$		
12	The integral controller enhances the				CO5	1
	a. Dynamic response	b. transient response	c. steady-state response	d. No effect of the controller		
Q. 2 Solve the following.						12
A)	The transfer function of a system is given below $G(s) = \frac{8(s+3)(s+4)}{s(s+2)^2 s^2+2s+5}$ Determine the poles and zeros and show pole-zero configuration in s-plane					6
B)	Compare open loop system with feedback control system.					6
Q.3 Solve the following.						12
A)	Find the transfer function of the system shown in figure by using block diagram reduction rules.					6

B)	Explain signal flow graph with the help of Masons gain formula.		6
Q. 4	Solve Any Two of the following.		12
A)	Determine the stability of a system having following characteristic equation $s^6+s^5+5s^4+3s^3+2s^2-4s-8$		6
B)	For the system having closed loop transfer function $\frac{C(s)}{R(s)} = \frac{18}{s^2 + 4s + 18}$ Determine i) Damped frequency of oscillations ii) Peak time iii) % peak overshoot		6
C)	The overall transfer function of a unity feedback control system is given by $\frac{C(s)}{R(s)} = \frac{10}{s^2 + 6s + 10}$ Determine i) step, ramp, parabolic coefficients ii) Steady state error if input is $r(t) = 1+t+t^2$		6
Q.5	Solve Any Two of the following.		12
A)	Plot the root locus of unity feedback control system has open loop transfer function $G(s) = \frac{K}{s(s+4)}$		6
B)	Enlist the different controllers. Write short note on PID controller.		6
C)	Describe the steps to construct Bode plot for any given system.		6
Q. 6	Solve Any Two of the following.		12
A)	Find state transition matrix for $\begin{bmatrix} 0 & -2 \\ 1 & -3 \end{bmatrix}$		6
B)	Explain the term Controllability and Observability with necessary equations.		6
C)	Obtain the state model for the system with transfer function $\frac{C(s)}{R(s)} = \frac{s^2 + 3s + 2}{s^3 + 9s^2 + 26s + 24}$		6
	*** End ***		

